Racing Genesis: Assignment 5 Beta Build Report & Production Roadmap

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Public GitHub Repository: https://github.com/VShaunak/RacingGenesis-2.0 | **Playable Build:** http://localhost:8081

1. Quick Review (1st Pillar: Procedural Modeling & Alpha Achievements)

Our primary technical pillar is **Procedural 3D Modeling & Continuous Track Ribbon Extrusion**. In Assignment 4 (Alpha Build), we solved the classical Frenet-Serret differential geometry singularity where straightaways (curvature $\kappa \to 0$) cause surface normal vectors to vanish or flip by 180°. We implemented a **Parallel Transport Frame (Bishop Frame)** algorithm anchored to a constant world-up vector, combined with periodic phantom ghost control points across the start/finish gantry to guarantee continuous tangent and second-derivative C^2 alignment. This framework procedurally extrudes the roadbed, curbs, gravel traps, safety barriers, and grandstands across 5 authentic Grand Prix circuits (Monza, Silverstone, Spa, Monaco, Suzuka) and user-sculpted circuits from 1,000m to 10,000m, running locked at 90 FPS WebGL.

2. Secondary Pillar Integration into the Core Driving Loop

For Assignment 5 (Beta Build), our core focus was expanding our vertical slice by directly connecting secondary technical pillars into the active driving loop, ensuring every feature interacts dynamically with the vehicle physics rather than running as an isolated tech demo:

- **Hardware-Instanced GPU Particle Shaders (Tire Smoke & Skidmarks):** Pre-allocated 120-particle THREE.InstancedMesh executing in a single draw call. Particles emit from tire contact patches whenever threshold braking (>38 m/s²) or lateral tire slip (>2.2G) occurs. Radial soft-alpha sprites expand and rise with thermal buoyancy while dynamic line segments deposit persistent skidmark decals along tire paths.
- **Brake Disc Thermal Incandescence (Molten Rotor Glow):** Kinetic energy dissipation ($E_k = 0.5mv^2$) calculates dynamic rotor heat up to 800°C. Under threshold deceleration, the carbon-ceramic material transitions from matte dark steel into molten incandescent amber (emissive #ff3300, intensity 2.8), cooling via convective airflow as speed increases down straightaways.
- **Multi-Car AI Racing Simulation (Sparring Opponent):** Instantiated a concurrent secondary Formula 1 car entity with Scarlet Red livery. The agent executes real-time spline lookahead pacing, anticipating upcoming corner radii, calculating apex speeds, and holding an overtaking lane offset with live HUD gap delta (+1.45s).
- **Curvilinear Boundary Collision & Curb Rumble Haptics:** High-frequency curb contact detection ($|d_lat| > roadWidth/2 - 0.7m$) triggers chassis micro-oscillations, camera shake, and 48 Hz Web Audio square-wave haptics; grass/gravel off-track incurs steep drag penalties (-34 m/s²).
- **Live Telemetry Canvas Oscillogram (90 Hz Multi-Channel Waveforms):** Real-time 75-sample rolling circular buffer in the HUD Telemetry drawer plotting synchronized traces for Speed (cyan), Throttle (green), Brake (red), and Lateral G-Force (amber).
- **Atmospheric Environment & Visual Cohesion:** Procedural daylight sky dome gradient texture blending warm horizon haze with alpine blue zenith, bridging the visual gap from programmer art into authentic Grand Prix aesthetics.

3. Beta New Features (Summary of Additions Since Alpha Build)

Feature Addition	Subsystem	Technical Implementation & Gameplay Impact
GPU Instanced Tire Smoke	Visual Particles	120-particle InstancedMesh pool emitted at tire contact patches under threshold braking/slip.
Dynamic Asphalt Skidmarks	Visual Shaders	Dynamic line-segment buffer geometry depositing rubber decals along tire cornering paths.
Brake Thermal Incandescence	Shader Dynamics	Thermal friction shader shifting rotors to glowing incandescent red (#ff3300), cooled by airspeed.
Multi-Car Sparring Opponent	AI & Simulation	Concurrent AI F1 car with Ferrari Scarlet livery, lookahead pacing, and real-time HUD gap delta.
Curb Rumble Strip Haptics	Audio & Physics	Chassis vertical shudder, camera screen shake, and 48 Hz synthesized Web Audio rumble buzzing.
Off-Track Drag Penalties	Boundary Physics	Severe grass/gravel deceleration penalty (-34 m/s²) and kicking dust sprites preventing corner-cutting.
Live Telemetry Oscillogram	HUD & Telemetry	90 Hz rolling canvas waveform graph plotting speed, throttle %, brake %, and lateral G curves.
Atmospheric Sky Gradient	Environment	Procedural horizon-to-zenith daylight color gradient replacing flat blue for visual cohesion.
Simulation Control Toggles	UI / Settings	In-drawer toggles for AI Sparring Car on/off and Particle FX on/off for instant benchmarking.

4. Known Issues & Unoptimized Areas (Heading into Buffer Week)

Heading into the final development weeks, we maintain transparent accounting of current prototype limitations:

- **Multi-Car Elastic Impulse Response:** While the opponent AI holds lane clearance and paces realistically, high-speed car-to-car collision currently relies on kinematic lateral separation rather than physical rigid-body collision meshes or elastic rebound impulses.
- **Web Audio Autoplay Policy:** Modern browser security gates suspend the AudioContext until the player's initial mouse interaction with the canvas or drawer UI controls.
- **Extreme Spline Self-Intersections:** If a user aggressively drags 3D handles into complex figure-8 knots, the extruded ribbons overlap co-planarly without generating elevation overpasses.
- **Touchscreen Calibration:** While fully responsive on mobile touch viewports, 3D handle raycasting is calibrated primarily for high-DPI desktop mouse pointers.

5. Production Schedule & Milestone Status (Week 9 to Week 14)

- **Week 9: Alpha Build (MVP 2.0) [COMPLETED]:** Complete 3-state game loop (Start → Active → Checkered Flag), 5 Grand Prix layouts, manual WASD + AI autopilot, 90 FPS rendering.
- **Week 10: Visual Shaders & Particles [COMPLETED]:** GPU instanced tire smoke, asphalt skidmarks, brake disc incandescence, procedural atmospheric sky.
- **Week 11: Boundaries & Curb Haptics [COMPLETED]:** Curb vibration, camera rumble, off-track drag penalties, audio harmonics.
- **Week 12: Multi-Car Simulation [COMPLETED]:** Sparring opponent AI, overtaking lane offset, dynamic pacing, live gap delta.
- **Week 13: Live Telemetry Pipeline [COMPLETED]:** 90 Hz rolling canvas waveform oscillogram, in-drawer module toggles.
- **Week 14: Dedicated Buffer Week (Feature Freeze) [UPCOMING]:** Strict feature freeze, cross-browser memory profiling (Chrome, Edge, Firefox), WebGL draw-call optimization, final video capture, and project documentation polish.